TUTORIAL SERIES · ML ENGINEERING

FROM ZERO TO KAGGLE GRANDMASTER

XGBoost
The Complete Visual Guide

A structured, from-scratch walkthrough of what XGBoost is, why it was built, how it achieves its speed and performance internally, and how to tune it like a professional — with diagrams, math, and practical rules of thumb at every step.

Basic → Intermediate → Advanced · Dark Mode Edition

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The guide is organized to build intuition first, then mechanics, then math, then tuning practice — in that order.

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How to read this guide: Every optimization is explained first with a plain-English analogy, then a diagram, and — only where it genuinely helps — the underlying math. You do not need to memorize equations to use XGBoost well, but understanding *why* each switch exists will make every hyperparameter make sense.

CHAPTER 01 · BASIC

Why Do We Even Need XGBoost?

To understand why XGBoost matters, we have to rewind to the basic question every ML beginner asks.

Machine Learning, in one sentence

Machine learning is a technique where a system **learns patterns from data** and uses those patterns to make predictions. You feed data into an algorithm, the algorithm learns the underlying structure, and then it predicts on new, unseen data.

The obvious beginner question: why so many algorithms?

If machine learning is just "learn from data," why do we need Linear Regression, Logistic Regression, Naive Bayes, KNN, SVM, Decision Trees, and dozens more? The answer: **algorithms are data-specific**. Each one is good at a narrow slice of problems.



Era 1 — the 1970s–80s: powerful but narrow

Early algorithms like Naive Bayes could genuinely solve problems using machine learning, but each one worked well only on a very specific type of data. Linear Regression assumes linear relationships. Naive Bayes shines mainly on text-like data. Step outside that comfort zone, and performance collapses.

Era 2 — the 1990s: powerful and general, but not scalable

The 1990s introduced three landmark algorithms: **Random Forest**, **SVM**, and **Gradient Boosting**. These were both strong performers *and* flexible across different data types. But they still had two real weaknesses:

1. Overfitting
They could still memorize noise in the training data instead of learning general patterns.

2. Scalability
As the internet exploded post-2000, datasets grew fast — and these algorithms slowed down badly on large data.

Era 3 — 2014: enter XGBoost

XGBoost was built specifically to solve those two remaining problems: **performance** (accuracy / generalization) and **speed** (scalability on large datasets). It didn't invent a new algorithm from scratch — it took an existing algorithm and re-engineered it aggressively.

COMMON MISCONCEPTION
XGBoost is **not** a brand-new machine learning algorithm. It is a highly optimized **library/implementation** built on top of the already-existing **Gradient Boosting** algorithm.

CHAPTER 02 · BASIC

The Origin Story

XGBoost was created by **Tianqi Chen** — the idea: take Gradient Boosting, layer heavy software-engineering optimizations on top, and turn a good algorithm into an extreme one.

THE FORMULA

XGBoost = Gradient Boosting (machine learning core) + Software engineering optimizations (speed & scale)

Three phases of history

1 Early Days — 2014

Tianqi Chen published the original paper introducing XGBoost as a highly scalable machine learning system. To prove it worked, he personally entered the **Higgs Boson Machine Learning Challenge** on Kaggle — a particle-physics competition — and **won it** using XGBoost. That win put the library on the map.

2 Uproar on Kaggle — 2016

Word spread fast. By 2016, out of **29 winning solutions** on Kaggle, **16 of them** used XGBoost somewhere in the pipeline. That statistic alone made the entire data science community take notice, and adoption snowballed.

3 Open Source Days — 2016 onward

Recognizing the momentum, Tianqi Chen open-sourced the project. From that point on, a global community of contributors added:

- Multi-platform and multi-language support
- Deeper optimizations for speed and accuracy
- Extensive documentation and tutorials

What might have taken 10 years of organic growth happened in roughly 5, thanks to open-source collaboration. By the 2020s, XGBoost had become close to a *default requirement* for anyone doing applied data science.



WORTH DOING

Read the original 2014 paper, "*XGBoost: A Scalable Tree Boosting System.*" This tutorial simplifies its key ideas, but the paper itself is short, well-written, and a great habit-builder for reading ML research.

CHAPTER 03 · BASIC

Why Gradient Boosting Was the Chosen One

Tianqi Chen had several strong algorithms available in the 90s-era toolbox. Here's specifically why Gradient Boosting won out as the base to build on.

Flexibility

Works with *any differentiable loss function* — regression, classification, ranking, even custom-defined objectives.

Performance

Consistently strong results across most datasets it's applied to.

Robustness

With proper regularization, results stay stable; it also copes reasonably with missing values internally.

Proven track record

Data scientists were already using vanilla Gradient Boosting to win Kaggle competitions — a strong signal that the underlying algorithm had untapped potential.

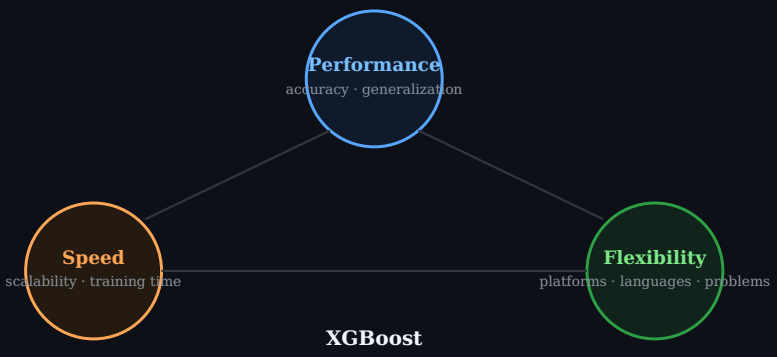
Putting it together: Gradient Boosting was already flexible, accurate, and robust. The only things missing were **raw training speed** and **scalability to huge datasets**. Fix those two, and you get something special. That gap is exactly what XGBoost was engineered to close.

> Gradient Boosting **+** Speed Engineering **+** Regularization & Scale Fixes **=** XGBoost
>
> The core thesis of the 2014 paper, in one line.

CHAPTER 04 · BASIC

The Three Pillars

Everything XGBoost does can be grouped under three design goals that Tianqi Chen prioritized from day one.



Pillar	What it means	Where we'll cover it
Flexibility	Runs anywhere, in any language, integrates with the rest of the ML ecosystem, solves almost any problem type.	Chapter 05
Speed	Six distinct software-engineering optimizations that make training dramatically faster on large data.	Chapter 06
Performance	Five machine-learning-level optimizations that push accuracy and reduce overfitting.	Chapter 07

Keep this three-way split in your head — almost every feature and hyperparameter you'll ever touch in XGBoost exists to serve one of these three goals.

CHAPTER 05 · INTERMEDIATE

Pillar 1 — Flexibility

Tianqi Chen's philosophy: a library only becomes great if it reaches as many people and use cases as possible. Four concrete design choices deliver on that.

1. Cross-platform

A model trained on Windows can run on Linux or Mac without modification. Table stakes today, but still worth naming as a deliberate design goal.

2. Multi-language support (the standout feature)

Most predictive libraries live inside one language — usually Python or R. XGBoost instead ships official wrappers/interfaces for **Python, R, C, C++, Java, Scala, Julia, Swift, Ruby**, and more. You can train a model in Python, save it, and load that exact model inside a Java application — genuinely useful when a company's production stack is Java-based but the data science work happens in Python.

```
# Python — train & save
model = xgb.train(params, dtrain)
model.save_model("model.json")

// Java — load the same model and predict
Booster booster = XGBoost.loadModel("model.json");
float[][] predictions = booster.predict(testMatrix);
```

WHY THIS MATTERS

Without this, deploying a Python-trained model inside a Java enterprise app normally means wrapping it as an API microservice — extra infrastructure, extra latency, extra failure points. Multi-language support removes that entire step.

3. Ecosystem compatibility

Data & modeling
NumPy · Pandas · scikit-learn · Matplotlib

Scale & deploy
Spark · Dask · Docker · Kubernetes

Explain & orchestrate
SHAP · LIME · Airflow · MLflow

4. Works across almost every ML problem type

Problem Type	Example
Regression	Predicting a continuous value (price, salary)
Binary classification	Spam vs. not-spam
Multi-class classification	Classifying an image into one of many categories
Time-series forecasting	Forecasting future demand or sales
Ranking	Recommender systems — ranking items by relevance
Anomaly detection	Fraud / outlier detection
Custom objectives	Any custom differentiable loss you design yourself

This last point traces back to Gradient Boosting's core trait: it can optimize *any differentiable loss function*. Want a loss that penalizes false positives more than false negatives? You can define that yourself.

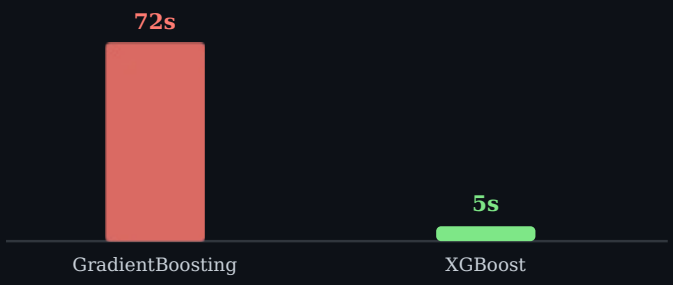
CHAPTER 06 · INTERMEDIATE

Pillar 2 — Speed

As datasets exploded in size, older algorithms simply became too slow. XGBoost applies six distinct engineering optimizations to solve this.

The proof, first

A simple benchmark: a synthetic dataset with 10,000 rows and 200 columns, one model trained with plain Gradient Boosting, one trained with XGBoost, same data.



~13–14× faster on the same data, same machine — and the gap widens further as data size grows.

The six speed optimizations, at a glance

#	Optimization	One-line idea
1	Parallel Processing	Evaluate multiple features simultaneously when building one tree
2	Column Block Storage	Store data column-wise, not row-wise, to enable that parallelism
3	Cache Awareness	Keep frequently-used values in fast CPU cache memory
4	Out-of-Core Computing	Train on datasets bigger than your RAM, in chunks
5	Distributed Computing	Split data and training across multiple machines
6	GPU Support	Offload histogram-building and split-finding to a GPU

1 Parallel Processing

THE NATURAL CONFUSION

Boosting is fundamentally **sequential** — tree 2 fixes tree 1's mistakes, tree 3 fixes tree 2's, and so on. So where does "parallel" fit in?

The parallelism isn't across trees — it's **inside the construction of a single tree**. Specifically, when deciding the best feature and threshold to split on at a node, XGBoost tests candidate features **simultaneously** instead of one after another.

Worked example

Suppose we're building a tree on two features — Age and Marks — to predict placement package.



With only two features, the time saved looks small. But real datasets often have **hundreds** of features. Testing them one-by-one versus testing 8 at a time (on an 8-core CPU) is the difference between minutes and seconds, repeated at *every single node* of *every single tree*.

HYPERPARAMETER

`n_jobs=-1` tells XGBoost to use all available CPU cores for this parallel feature evaluation.

2 Column Block Storage

Parallel feature evaluation above is only possible because of *how* XGBoost stores data internally. Most tools store data **row-wise** (process one full row, then the next). XGBoost instead stores data in **Column Blocks** — each feature's values are kept together, contiguous in memory.



When it's time to parallelize split-finding, each CPU core can grab a whole column block and work on it independently — no need to scan across unrelated rows to isolate one feature's values. This is a pure data structure decision, but it's what unlocks optimization #1.

3 Cache Awareness

Every computer has a CPU (does the math) and RAM (holds the data). They talk over a system bus, and that communication takes time. **Cache memory** is a small, very fast memory sitting inside the CPU itself (typically a few MB) — used for anything accessed repeatedly.

KITCHEN ANALOGY

RAM is the **fridge** across the kitchen. CPU is the **cook**. Cache memory is the **small tray of ingredients kept right next to the cook** because they're needed over and over — no more trips to the fridge for those items.

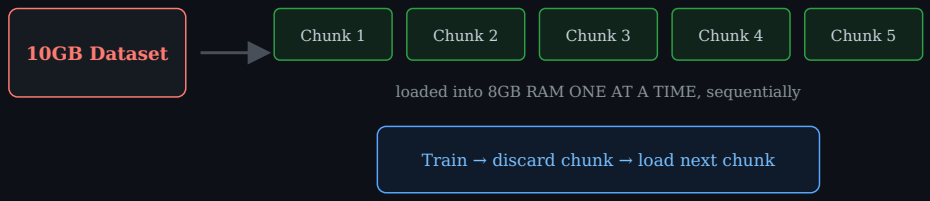


XGBoost builds trees using **histogram-based training**: numeric features are bucketed into bins (like a histogram), and those bin boundaries and statistics get reused constantly while building a tree. XGBoost deliberately keeps this histogram data in cache memory, so the CPU doesn't repeatedly wait on slower RAM round-trips. The net effect: noticeably faster training.

4 Out-of-Core Computing

What happens when your dataset is **bigger than your RAM**? Say your laptop has 8GB RAM but the dataset is 10GB — you can't even load it, let alone train on it.

Out-of-core computing solves this by splitting the large dataset into smaller chunks, loading one chunk into RAM at a time, training on it, discarding it, then loading the next.



HYPERPARAMETER

Setting `tree_method='hist'` engages the histogram-based / out-of-core-friendly training path, letting you work with datasets far larger than a single machine's RAM.

Cache awareness and out-of-core computing work together: as each chunk streams through, the frequently-reused histogram statistics still get cached for speed.

5 Distributed Computing

Out-of-core computing still uses **one machine**, processing chunks sequentially. Distributed computing goes further: spread the chunks across **multiple machines (nodes)** so they train *in parallel*, not in sequence.



The three steps

1. **Partition** — split the dataset evenly across N nodes
2. **Train locally** — each node independently finds its best split candidates
3. **Aggregate** — a master node collects every candidate, compares the resulting gain, and selects the winner

VS. OUT-OF-CORE COMPUTING

Out-of-core = 1 machine, chunks processed one after another (sequential).
Distributed = N machines, chunks processed at the same time (parallel) → both handles bigger-than-RAM data *and* trains faster.

TOOLING

XGBoost doesn't implement distributed orchestration itself — it integrates with external frameworks like `Dask` or `Kubernetes` to manage the nodes and communication.

6 GPU Support

A CPU has a few very powerful cores. A **GPU** has thousands of small, less-powerful cores — perfect for many small, repeated calculations done in parallel. Since histogram-building and split-finding in XGBoost are *highly parallelizable* tasks, they map naturally onto a GPU.



The GPU curve reaches low error dramatically faster — in benchmark charts from the XGBoost team, GPU training reaches strong accuracy in a fraction of the time a 32-core CPU needs.

HYPERPARAMETER
Setting `tree_method='gpu_hist'` tells XGBoost to use the GPU (if available) for histogram construction and split finding — usually a drop-in change, no other code rewrite required.

Why "Extreme"?

Put all six speed optimizations together with the five performance optimizations in the next chapter, and you get a picture of just how aggressively engineered this library is — parallelism, custom data structures, cache tricks, out-of-core streaming, distributed training, and GPU acceleration, all stacked on top of one algorithm. That's exactly why it's called **Extreme Gradient Boosting**.

CHAPTER 07 · ADVANCED

Pillar 3 — Performance

Speed alone doesn't help if the model isn't accurate. This chapter covers the five machine-learning-level optimizations that make XGBoost generalize better than vanilla Gradient Boosting.

#	Optimization	Solves
1	Regularized Learning Objective	Overfitting, built directly into the loss function
2	Sparsity-Aware Split Finding	Missing values, without manual imputation
3	Approximate Tree Learning	Slow exact split search on huge feature sets
4	Weighted Quantile Sketch	Poor-quality, data-blind binning
5	Tree Pruning (with gamma)	Overly complex, deep, overfit trees

1 Regularized Learning Objective

Regularization is the standard fix when a model performs well on training data but poorly on test data (overfitting) — you add a penalty term to the loss function that discourages excessive complexity.

Vanilla Gradient Boosting only regularized *indirectly*: through the learning rate (shrinking each tree's contribution) and through pruning during tree construction. XGBoost goes further and bakes a regularization term **directly into the loss function itself**.

$$\text{Obj}(\theta) = \Sigma_i \, l(y_i, \hat{y}_i) \; + \; \Sigma_k \, \Omega(f_k)$$

$l(y_i, \hat{y}_i)$ = differentiable training loss (how wrong the prediction is) · $\Omega(f_k)$ = regularization term for tree f_k, penalizing complexity

$$\Omega(f) = \gamma T \; + \; \tfrac{1}{2} \, \lambda \, \Sigma_j \, w_j^2$$

T = number of leaves in the tree · w_j = weight (score) of leaf j · γ, λ = regularization hyperparameters (gamma, reg_lambda)

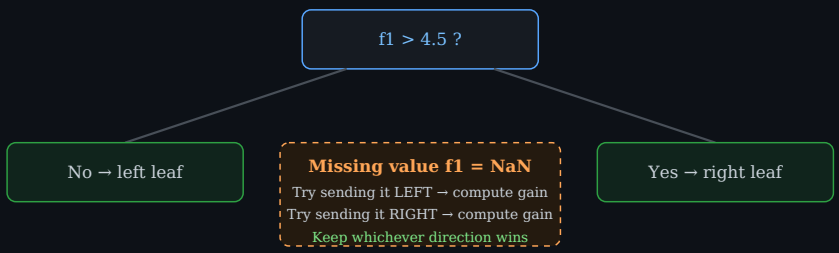
Because this penalty is part of the objective being minimized, regularization happens **automatically as a side-effect of training** — not as an afterthought bolted on after the tree is built. The result: models that generalize better by default, across a wider range of datasets.

2 Sparsity-Aware Split Finding & Missing Values

Sparsity means lots of zeros or missing values in your data. Every other classic algorithm requires you to impute (fill) or drop missing values before training. XGBoost handles them natively.

How it decides where a missing value belongs

At every split, XGBoost doesn't guess — it **tests both directions** and keeps whichever gives higher gain.



XGBoost effectively learns a **default direction** for missing values at every node, based on which direction maximizes the split's gain. This is what "sparsity-aware" means — the algorithm is explicitly aware of where the gaps and zeros are, and routes them intelligently rather than crashing or requiring pre-processing.

PRACTICAL EFFECT

You can pass a NumPy array or DataFrame with real `NaN` values straight into `XGBClassifier` / `XGBRegressor` — no imputation step required. Most other classic ML algorithms will simply error out on NaNs.

3 Efficient Split Finding

Exact Greedy Search (the slow, precise way)

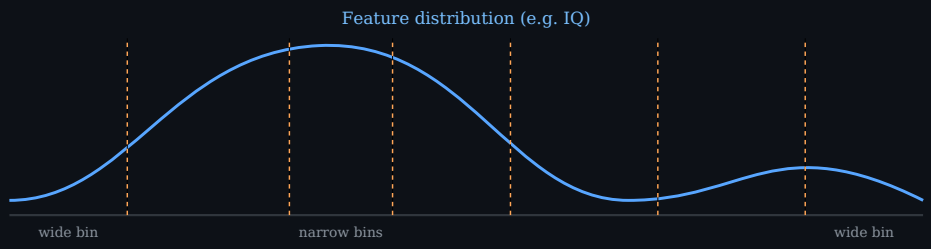
The textbook way to find the best split: sort every numeric feature, try the midpoint between *every* pair of consecutive values as a candidate threshold, and measure the resulting information gain for each candidate. It gives the best possible split — but is painfully slow on large datasets (imagine doing this for 10 million rows).

Approximate Tree Learning (the fast way)

Instead of trying every possible value, XGBoost **bins** each numeric feature into a limited number of buckets first, then only tests the bucket boundaries as candidate splits. Continuous becomes discrete — you trade a small amount of split precision for a large speed gain. This binning process is what we call **Histogram-Based Training**.

Weighted Quantile Sketch — how the bins are actually chosen

The naive approach would be *uniform* binning (e.g., always split 0–100 into ten equal chunks of 10). But that ignores the actual shape of your data. XGBoost instead bins based on **quantiles of the data's distribution** — so where data points are dense, bins are narrow (more resolution); where data is sparse, bins are wide.



Where the curve is dense (lots of data points), bins are narrow — giving the model finer resolution exactly where it's needed. Where the curve is sparse, wide bins are fine since there's little information to lose. The result: histogram bins that actually respect the data's shape, recovering much of the accuracy that plain uniform binning would have thrown away — while keeping the speed advantage of Approximate Tree Learning.

HYPERPARAMETERS

`tree_method='hist'` activates histogram-based (approximate) training. `max_bin` controls how many buckets are created per feature (default 256).

4 Tree Pruning

Pruning means trimming a tree's depth/complexity to reduce overfitting. XGBoost supports both classic strategies, plus one extra tool that vanilla Gradient Boosting lacks.

Pre-pruning

Decide the tree's size limits *before/during* construction (e.g. `max_depth`, `min_child_weight`) — stop growing branches that don't meet the criteria.

Post-pruning

Let the tree grow fully first, then trim branches back afterward based on whether they actually helped.

The gamma parameter — an ROI test for every split

XGBoost adds a unique lever: `gamma` (also called `min_split_loss`). A new branch is only created if it reduces the loss by **at least gamma**. Otherwise, that split is pruned away — even if it was technically possible to make.



This is the last of the five performance optimizations. Combined with speed optimizations from Chapter 6, you can now see the full picture: heavy software engineering (speed) plus deliberate statistical safeguards (performance), stacked on the flexible base of Gradient Boosting.

CHAPTER 08 · INTERMEDIATE

Why "Extreme"? Meet the Cousins

Put together: parallel processing, column-block storage, cache awareness, out-of-core computing, distributed computing, GPU support, regularized objectives, sparsity-awareness, approximate + weighted-quantile split finding, and gamma-based pruning. That is an unusually long list of aggressive optimizations layered onto one base algorithm — which is exactly why the creators named it **Extreme Gradient Boosting**.

XGBoost isn't the only strong Gradient Boosting implementation

Library	Built by	Known for
XGBoost	Tianqi Chen / DMLC	The original extreme-optimization pioneer; huge ecosystem & language support
LightGBM	Microsoft Research	Often faster training, lower memory usage; leaf-wise tree growth; strong at scale
CatBoost	Yandex	Best-in-class native handling of categorical features, no manual encoding needed

All three are Gradient Boosting Decision Tree (GBDT) implementations, and all three support parallel, distributed, and GPU training. Results between them are often close — sometimes LightGBM or CatBoost even edges out XGBoost, depending on the dataset. Knowing all three, and when each tends to shine, is a hallmark of a strong applied ML engineer.

CHAPTER 09 · ADVANCED

Hyperparameter Field Guide

Now that you know *why* each mechanism exists, here's the practical dial for each one.

Group A — Boosting Process

Parameter	Plain English	Typical Range
`n_estimators`	Number of sequential trees (relay-race team size)	Set high (2000–5000+) + early stopping
`learning_rate` / `eta`	How aggressively each tree corrects prior mistakes	0.01 – 0.05 (low & steady wins)

Group B — Tree Complexity

Parameter	Plain English	Typical Range
`max_depth`	How deep a single tree can grow	3 – 8
`min_child_weight`	Minimum "evidence" (hessian weight) needed to justify a new split	1 – 10
`gamma` / `min_split_loss`	Minimum loss reduction required to allow a split — an ROI hurdle	0 – 5, tune later

Group C — Randomness / Regularization

Parameter	Plain English	Typical Range
`subsample`	Fraction of rows randomly used per tree (row blinders)	0.6 – 0.9
`colsample_bytree`	Fraction of columns randomly used per tree (column blinders)	0.6 – 0.9
`reg_alpha` (L1)	Pushes unimportant feature weights to exactly zero	1e-5 – 100, for wide/sparse data
`reg_lambda` (L2)	Smooths leaf weights, prevents any single leaf dominating	Default 1, fine-tune last

Group D — Special-Case Parameters

Parameter	When to use it
`scale_pos_weight`	Heavily imbalanced targets — set to count(negative) / count(positive)
`tree_method='hist'`	Large datasets (100k+ rows) — 5–10× training speedup
`monotone_constraints`	Regulated domains (finance) — force predictions to respect known real-world direction
`max_bin`	Used with `tree_method='hist'`; lower it (e.g. 64) as a regularizer if overfitting

CHAPTER 10 · PRO

The Kaggle Grandmaster Tuning Workflow

The Core 5 — tune these first, tune these most

In practice, a small set of parameters delivers roughly 95% of the achievable performance. This is the daily-driver stack every experienced practitioner reaches for first.

Order	Parameters	Strategy
1	`learning_rate` + `n_estimators`	Fix learning_rate low (0.01–0.05), set n_estimators high, use `early_stopping_rounds`
2	`max_depth`	Search 3–7; deep trees overfit fast in XGBoost
3	`min_child_weight`	Search 1–10; your main overfitting brake
4	`subsample`	Search 0.6–0.9; cheap, effective regularization
5	`colsample_bytree`	Search 0.6–0.9; stops over-reliance on one strong feature

Staged tuning order (the winning sequence)

1. **Fix** a high `n_estimators` (~5000) and a low `learning_rate` (~0.05)
2. **Stage 1 – Complexity:** tune `max_depth` (3–9) and `min_child_weight` (1–10)
3. **Stage 2 – Randomness:** tune `subsample` and `colsample_bytree` (0.6–1.0)
4. **Stage 3 – Regularization:** tune `gamma`, `reg_alpha`, `reg_lambda`
5. **Stage 4 – Final polish:** drop `learning_rate` to ~0.01, raise `n_estimators` to ~10000, run once more with early stopping

"In case of emergency" parameters — only add these if you observe the specific problem

Symptom	Add	Value
Target is heavily imbalanced (e.g. 99% / 1%)	`scale_pos_weight`	count(negative) / count(positive), set statically — don't sweep it
Hundreds of sparse / noisy columns	`reg_alpha`	Search [1e-5, 1e-2, 0.1, 1, 10, 100]
Training is too slow (100k+ rows)	`tree_method`	Set statically to `'hist'`
Still overfitting despite low `max_depth`	`gamma`	Search 0.1 – 5.0

GRID SEARCH IS DEAD

Don't use `GridSearchCV` — testing every combination wastes compute. Bayesian search tools like **Optuna** or **Hyperopt** learn from previous trials and explore promising regions of the search space far more efficiently. ~30–50 trials on the Core 5 is usually enough.

CHAPTER 11 · PRO

Cheat Sheet & Final Takeaways

Golden rules from experienced practitioners

The 90/10 rule
Feature engineering, data cleaning, and validation strategy drive ~90% of model performance. Hyperparameter tuning squeezes out the last ~10%. Don't spend a week tuning a model built on bad data.

Strict data isolation
The `eval_set` used for early stopping must be completely separate from training data — any leakage makes early stopping halt at the wrong point, giving false confidence.

Fold stability > peak metric
A model scoring 0.85 consistently across 5 CV folds beats one scoring 0.89 on fold 1 but 0.72 on fold 3. High variance across folds means your hyperparameters are memorizing specific splits.

Seed averaging
Once you have your best hyperparameters, retrain 5 times with 5 different random seeds and average the predictions. Smooths out edge-case errors, almost always a small final boost.

IMBALANCED DATA TRAP
If only 2% of your data is positive, a model that always predicts "negative" is already 98% accurate — but useless. Never judge imbalanced problems with plain Accuracy or standard AUC. Use `aucpr` (Area Under the Precision-Recall Curve) or a custom F1-based metric instead.

WATCH THE DELTA, NOT JUST THE METRIC
Plot training loss vs. validation loss. If validation improves by 0.01 but the train/validation gap doubles, the model is becoming brittle — increase `min_child_weight` or reduce `max_depth`.

The three pillars, one last time

Pillar	Core idea	Key levers
Flexibility	Runs anywhere, any language, any problem type	objective, multi-language APIs, ecosystem integrations
Speed	Parallelism, smart data structures, cache, distribution, GPU	n_jobs, tree_method, gpu_hist
Performance	Regularized objective, sparsity-aware splits, smart binning, pruning	reg_alpha/lambda, gamma, max_bin, min_child_weight

WHERE TO GO NEXT
You now have the full map: what XGBoost is, why it was built, how every major optimization works internally, and how to tune it like a professional. The natural next steps are hands-on: run the staged tuning workflow on a real dataset, read the original 2014 paper for the full math, and compare results against LightGBM and CatBoost on the same data.